

JSS 3 MATHEMATICS – FIRST TERM

TOPIC 2: WHOLE NUMBERS II

1. Expressions Involving Brackets and Fractions

Work from the inside of brackets outwards. For fractions, find a common denominator when adding or subtracting; multiply numerators and denominators when multiplying; invert and multiply when dividing. Always simplify where possible.

2. Direct and Inverse Proportion

Direct proportion: as one quantity increases, the other increases at the same rate ($y = kx$). **Inverse proportion:** as one increases, the other decreases ($y = k/x$). Find the constant k from given values, then solve for unknowns.

3. Applications

Direct: cost and quantity at fixed price; distance and time at constant speed. Inverse: more workers, less time for the same job (under ideal conditions); pressure and volume of a gas (simplified model).

4. Compound Interest (Introduction)

Compound interest is interest calculated on the principal plus previously earned interest. A basic formula is $A = P(1 + r/100)^n$ where A is amount, P principal, r rate per period, n number of periods. Simple interest is only on the original principal.

5. Summary

Handle brackets and fractions carefully; use proportion for related quantities; compound interest grows the amount by adding interest to interest.

End of Topic 2